

Architecture of a Brain-Inspired Quantum Supercomputer and its Separation From Consciousness

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Abstract

In this note, the authors propose a quantum supercomputer. We propose to base our architecture on the multi-computer supercomputer architecture, Kitaev's quantum computation model and [3, 4]. For the basic electronics we propose a quantum version of operational amplifier axons. Finally, we propose to measure its quantum-phi quantity so as to measure its distance from genuine consciousness.

In [2] the authors considered creating devices that are cosmic conscious or even conscious as a distant goal. In [1] the authors proposed that neural systems could be imitated via op-amps on an electronics substrate. In this work the authors propose an architecture for a quantum supercomputer that mimics the brain and therefore a priori appears as a good candidate for a conscious device. To build such a device, the problems we need to solve are listed:

1. Develop a Quantum Model of Operation Amplifier Axons
2. Implement Field-Based Control of Multiple Quantum Axons
3. Incorporate Kitaev's Model of a Quantum Computer in the Proposed Axonal Architecture
4. Use the Multi-Computer Configuration for Networked Supercomputing
5. Derive the Quantum Version of Phi Based on Tononi's Integrated Information Theory
6. Find the Quantum-Phi of the Proposed Architecture

A quantum supercomputer modeled on the brain has perhaps the best possible chance of turning out to be conscious in the I.I.T. sense - it would be integrated with physical reality in the full sense and at the same time its high level complex design would mimic the brain. However, technologically, such a device is perhaps about a century away, given that full scale quantum computers are themselves

about half a century away as per well-known physicists. Defining and measuring its “cosmic” consciousness is farther away, and would be based on the quiescence of synchronization, which would enable subtle universal information inflow [5].

References

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